Akoizo Original Practice - 2027 Rotation Form 3 - Answer key

Division C | Chemistry Lab | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) At constant temperature, a gas at 1.6 atm occupies 7 L. Compress it to 3.5 L. Find the final pressure.

answer: 3.2 atm (2 points).

Numerical target 3.2; absolute tolerance 0.0064. Accepted units: , atm.

2. (2 points) At constant pressure, a gas occupies 7 L at 360 K. Find its volume at 720 K.

answer: 14 L (2 points).

Numerical target 14; absolute tolerance 0.028. Accepted units: , L.

3. (2 points) A rigid vessel contains a gas at 1.6 atm and 360 K. Find its pressure at 180 K with no gas lost.

answer: 0.8 atm (2 points).

Numerical target 0.8; absolute tolerance 0.0016. Accepted units: , atm.

4. (2 points) Use PV=nRT with R=0.08206 L atm/(mol K). P=1.6 atm, V=7 L and T=360 K. Find the amount of gas.

answer: 0.3791263844883148 mol (2 points).

Numerical target 0.37912638; absolute tolerance 0.00075825277. Accepted units: , mol.

5. (2 points) A dry gas mixture has partial pressures 1.6 atm and 0.8 atm. Find total pressure.

answer: 2.4000000000000004 atm (2 points).

Numerical target 2.4; absolute tolerance 0.0048. Accepted units: , atm.

6. (2 points) A mixture has 5 mol oxygen and 15 mol nitrogen at total pressure 7 atm. Find oxygen partial pressure.

answer: 1.75 atm (2 points).

Numerical target 1.75; absolute tolerance 0.0035. Accepted units: , atm.

7. (2 points) Gas collected over water has total pressure 103 kPa. Water-vapor pressure is supplied as 6 kPa. Find dry-gas pressure.

answer: 97 kPa (2 points).

Numerical target 97; absolute tolerance 0.194. Accepted units: , kPa.

8. (2 points) Two ideal gases at the same temperature have molar masses 4 and 16 g/mol. What is the diffusion-rate ratio light/heavy? Use Graham's law.

answer: 2 (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , .

9. (2 points) For 2 H2 + O2 -> 2 H2O, 5 mol O2 reacts completely with excess H2. How many moles of H2 are consumed?

answer: 10 mol (2 points).

Numerical target 10; absolute tolerance 0.02. Accepted units: , mol.

10. (2 points) Explain why gas-law temperature ratios must use kelvin rather than degrees Celsius.

answer: Kelvin is proportional to absolute thermodynamic temperature. Celsius has an arbitrary offset; its ratios do not represent molecular kinetic-energy ratios. (2 points).

11. (2 points) [State/National] Determine the order with respect to A from the supplied initial-rate data.

Synthetic initial-rate data (M/s): [A]=0.16 M, [B]=0.10 M gives rate 0.0128; [A]=0.32, [B]=0.10 gives 0.0512; [A]=0.16, [B]=0.20 gives 0.0256. Assume rate=k[A]^m[B]^n.

answer: 2 (2 points).

Numerical target 2; absolute tolerance 0.004. Accepted units: , .

12. (2 points) [State/National] Determine the order with respect to B from the supplied initial-rate data.

Synthetic initial-rate data (M/s): [A]=0.16 M, [B]=0.10 M gives rate 0.0128; [A]=0.32, [B]=0.10 gives 0.0512; [A]=0.16, [B]=0.20 gives 0.0256. Assume rate=k[A]^m[B]^n.

answer: 1 (2 points).

Numerical target 1; absolute tolerance 0.002. Accepted units: , .

13. (2 points) [State/National] Calculate k from the initial-rate data. Give its value in M^-2 s^-1.

Synthetic initial-rate data (M/s): [A]=0.16 M, [B]=0.10 M gives rate 0.0128; [A]=0.32, [B]=0.10 gives 0.0512; [A]=0.16, [B]=0.20 gives 0.0256. Assume rate=k[A]^m[B]^n.

answer: 5 M^-2 s^-1 (2 points).

Numerical target 5; absolute tolerance 0.01. Accepted units: , M^-2 s^-1.

14. (2 points) [State/National] Predict the rate at [A]=0.48 M and [B]=0.10 M.

Synthetic initial-rate data (M/s): [A]=0.16 M, [B]=0.10 M gives rate 0.0128; [A]=0.32, [B]=0.10 gives 0.0512; [A]=0.16, [B]=0.20 gives 0.0256. Assume rate=k[A]^m[B]^n.

answer: 0.11520000000000002 M/s (2 points).

Numerical target 0.1152; absolute tolerance 0.0002304. Accepted units: , M/s.

15. (2 points) [State/National] For a first-order reaction, k=0.05330769 s^-1. Use ln(2)=0.693 to calculate the half-life.

answer: 13 s (2 points).

Numerical target 13; absolute tolerance 0.026. Accepted units: , s.

16. (2 points) [State/National] A first-order reactant starts at 1.04 M. Find concentration after three half-lives.

answer: 0.13 M (2 points).

Numerical target 0.13; absolute tolerance 0.00026. Accepted units: , M.

17. (2 points) [State/National] A zero-order reaction starts at 1.3 M and consumes reactant at 0.02 M/s. Find concentration after 10 s.

answer: 1.1 M (2 points).

Numerical target 1.1; absolute tolerance 0.0022. Accepted units: , M.

18. (2 points) A catalyst increases rate without changing reactant/product free energies. Explain the mechanism and whether the equilibrium constant changes.

answer: It offers a lower-activation-energy pathway and speeds approach to equilibrium. At fixed temperature it does not change the equilibrium constant. (2 points).

19. (2 points) Equal masses of the same solid react with excess acid under matched conditions. Why can smaller particles react faster?

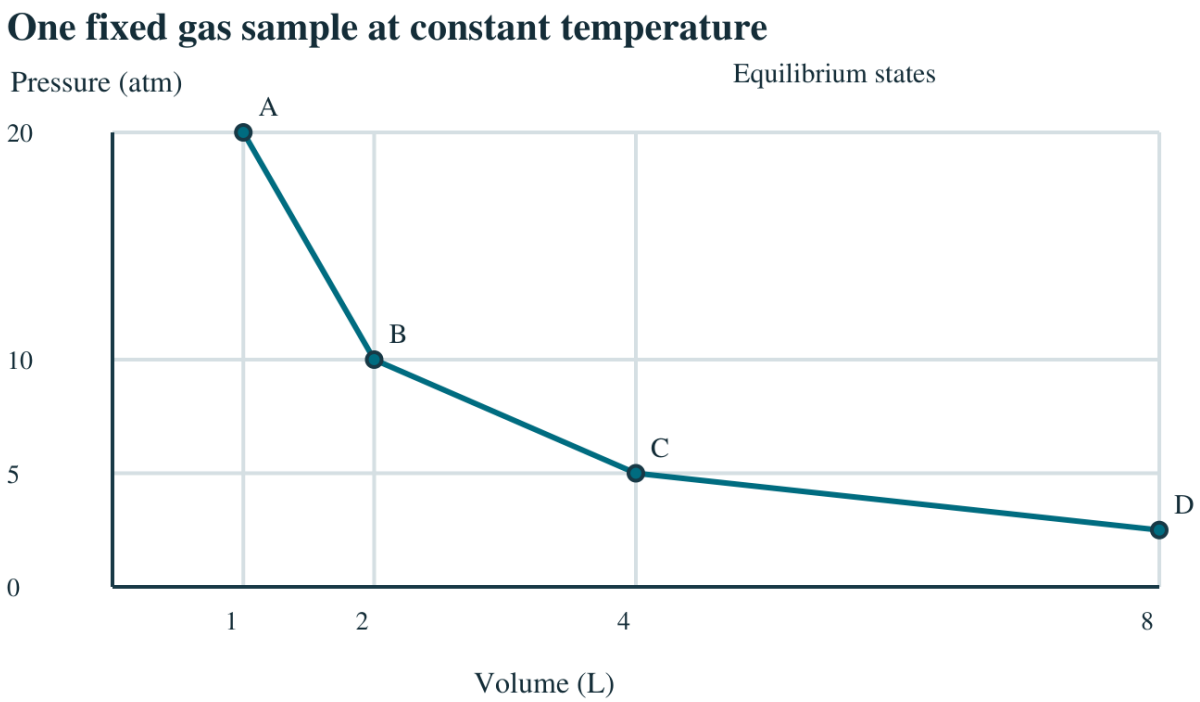
answer: Smaller particles expose more surface area, increasing contact and collision opportunities; mass alone does not set exposed area. (2 points).

20. (2 points) [State/National] At 300 K a rate constant is k1; at 330 K it is 2k1. Using R=8.314 J/(mol K), ln(2)=0.693 and ln(k2/k1)=Ea/R(1/T1-1/T2), find Ea.

answer: 19013.286599999985 J/mol (2 points).

Numerical target 19013.287; absolute tolerance 38.026573. Accepted units: , J/mol.

21. (2 points) Use Figure F1. Read the pressure at state C.



F1. Ideal gas isotherm (original synthetic teaching figure)

answer: 5 atm (2 points).

Numerical target 5; absolute tolerance 0.015. Accepted units: , atm.

22. (2 points) Use Figure F1. Calculate PV at B.

Refer to F1. Ideal gas isotherm (original synthetic teaching figure) shown with the first question in this figure group.

answer: 20 L atm (2 points).

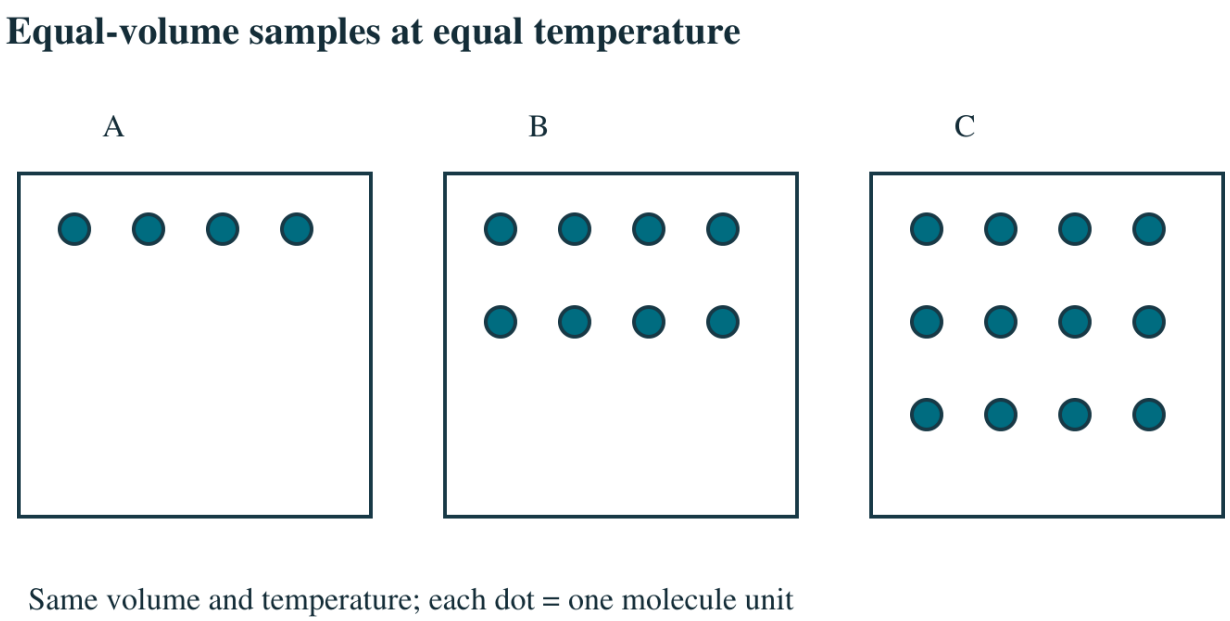
Numerical target 20; absolute tolerance 0.06. Accepted units: , L atm.

23. (2 points) Use Figure F1. Use two plotted states to explain the volume-pressure relationship.

Refer to F1. Ideal gas isotherm (original synthetic teaching figure) shown with the first question in this figure group.

answer: For example, volume doubles A to B while pressure halves; PV remains constant for this fixed amount at constant temperature. (2 points).

24. (2 points) Use Figure F2. Find the pressure ratio B/A.



F2. Ideal gas particle models (original synthetic teaching figure)

answer: 2 (2 points).

Numerical target 2; absolute tolerance 0.006. Accepted units: .

25. (2 points) Use Figure F2. If A contains 0.10 mol, how many moles does C contain?

Refer to F2. Ideal gas particle models (original synthetic teaching figure) shown with the first question in this figure group.

answer: 0.3 mol (2 points).

Numerical target 0.3; absolute tolerance 0.0009. Accepted units: , mol.

26. (2 points) Use Figure F2. Would the dots justify a claim that C has faster molecules than A? Explain.

Refer to F2. Ideal gas particle models (original synthetic teaching figure) shown with the first question in this figure group.

answer: No; equal temperature fixes the ideal-gas average translational kinetic energy, while this picture varies number density. Speed also depends on molecular mass. (2 points).

27. (2 points) Use Figure F3. [State/National] Determine reaction order with respect to A from runs 1 and 2.

Matched-temperature initial-rate measurements

Run	[A] (M)	[B] (M)	Rate (M/s)
1	0.10	0.10	0.005
2	0.20	0.10	0.020
3	0.10	0.20	0.010

F3. Initial-rate experiment (original synthetic teaching figure)

answer: 2 (2 points).

Numerical target 2; absolute tolerance 0.006. Accepted units: .

28. (2 points) Use Figure F3. [State/National] Determine reaction order with respect to B from runs 1 and 3.

Refer to F3. Initial-rate experiment (original synthetic teaching figure) shown with the first question in this figure group.

answer: 1 (2 points).

Numerical target 1; absolute tolerance 0.003. Accepted units: .

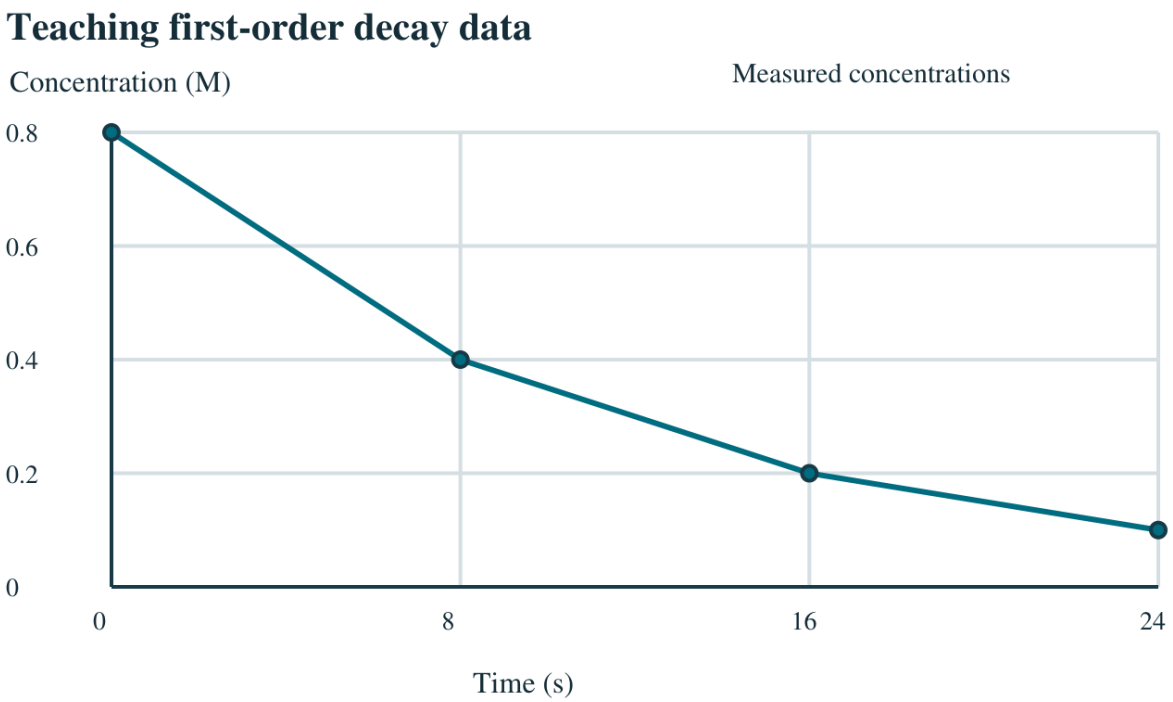
29. (2 points) Use Figure F3. [State/National] Calculate k for rate=k[A]^2[B].

Refer to F3. Initial-rate experiment (original synthetic teaching figure) shown with the first question in this figure group.

answer: 5 M^-2 s^-1 (2 points).

Numerical target 5; absolute tolerance 0.015. Accepted units: , M^-2 s^-1.

30. (2 points) Use Figure F4. [State/National] Read the half-life.



F4. Reactant decay curve (original synthetic teaching figure)

answer: 8 s (2 points).

Numerical target 8; absolute tolerance 0.024. Accepted units: , s.

31. (2 points) Use Figure F4. [State/National] Use k=ln(2)/half-life with ln(2)=0.693 to calculate k.

Refer to F4. Reactant decay curve (original synthetic teaching figure) shown with the first question in this figure group.

answer: 0.086625 s^-1 (2 points).

Numerical target 0.086625; absolute tolerance 0.000259875. Accepted units: , s^-1.

32. (2 points) Use Figure F4. [State/National] Explain why the equal time intervals do not remove an equal concentration each time.

Refer to F4. Reactant decay curve (original synthetic teaching figure) shown with the first question in this figure group.

answer: For first-order decay an equal fraction remains after each half-life; the absolute concentration decrement becomes smaller as concentration falls. (2 points).